Natural Language Processing
&
Ontology

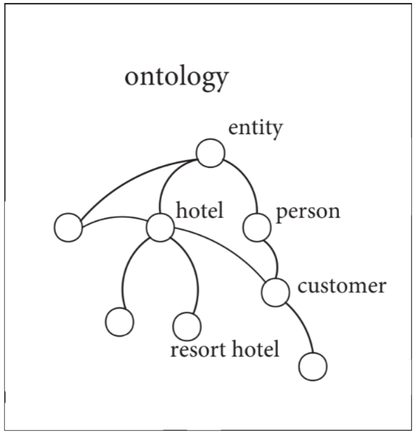
Amitesh Kumar , Viswas Paul Johny
H00MACL202500120 H00MACL202500114

Introduction

- Ontology refers to the study of the meaning of existence, and the relationship of said entities with other entities. At the core, it is the study of meaning itself.
- Formally, ontology is said to be “the formal specification of a shared conceptualisation”
- In Computer Science, and in NLP, ontology is the “formal and structured representation of knowledge.
- It helps machines understand meanings rather than just words, entities rather that concepts, and relationships rather than islets of ideas.
- As of now, ontologies are far past what they once were expected to evolve into, being now but “networked fully structured knowledge models with concepts, relations, rules/axioms”
- Our continual usage of instances of NLP, including but not limited to the likes of Claude, ChatGPT, Mistral, are key factors in the contribution to any domain specific Ontology.

NLP & Ontology

- In NLP, ontology connects language and meaning.
- It defines how words, phrases, and entities are related conceptually.
- Example: Cat → Mammal → Animal | Dog → Mammal → Animal
- ‘Cat chases dog’ = interaction between animals.
- These relationships can be directed or bidirectional.
- Ontologies, which will be explained later, are usually referred to in CompSci and in NLP application using the equation G=(EV), where E refers to edges, that in turn refer to entities and V to verticies that refer to relationships between edges.



Components of Ontology

- Concepts (Classes): Entities or ideas (Animal, Person, City)
- Instances: Actual examples (Dog, Cat, Paris)
- Relations: Connections (Dog is a Animal)
- Properties: Attributes (hasColor, livesIn)
- Hierarchy: General → Specific (Animal → Mammal → Dog)
- However, before an ontology is made, it however has to go through the following path before coming into being.

unstructured text → terminology → glossary → thesaurus → taxonomy → ontology

- **Unstructured text** is text as is, **terminology**, words that appear in the text and are of importance to the domain, **glossary**, a list of terminology against their dictionary meaning, **thesaurus**, glossary + synonyms, **taxonomy**, entities within the glossary in an hierarchical manner, and **ontology**, finally brings about relationships within and outside the ontology,

Ontologies : Dissected Esoterically

- Concepts (Classes): The basic units of an ontology; abstract categories or meanings.
- Instances (Individuals): Concrete entities (ABox assertions) that belong to concepts.
- Relations: Connect concepts and instances. Common relation types include taxonomic (is-hierarchies), hypernyms, and partonomic (part-of) links, meronymy.
- Attributes/Properties: Datatype or object properties that give concepts specific attributes (e.g., hasColor, hasName). These refine concept definitions.
- Axioms and Constraints: Logical statements (e.g. disjointness, cardinality, domain/range restrictions) that enforce consistency. E.g., declaring Male and Female as disjoint classes ensures no instance can be both .
- Rules: (Optional) Complex logical rules (e.g. SWRL [extended OWL]) can infer new facts. Ontologies may include rules or integrate with rule engines for advanced inference.
- Terminology Box (TBox) vs Assertion Box (ABox): In formal languages (e.g. OWL or SWRL, which is used in CompSci for ontologies for Semantic Web) TBox contains class definitions (concepts/properties) and ABox contains instance assertions.

```
### Define classes
Class: Animal

Class: Bird
    SubClassOf:
        Animal,
        hasWings exactly 2,
        hasAbility some Fly

Class: Fly
    SubClassOf: Ability

Class: Swim
    SubClassOf: Ability

Class: Penguin
    SubClassOf:
        Bird,
        hasAbility only Swim

### Define properties
ObjectProperty: hasWings
    Domain: Bird
    Range: Wing

ObjectProperty: hasAbility
    Domain: Animal
    Range: Ability
```

This is how the class bird would be defined with seemingly less obvious logical or real-life contradictions.

Written using Manchester OWL, a dialect of OWL, used in Protégé, an IDE for Ontology

Cooking up Ontologies

- Ontology Building: Manual or semi-automatic engineering of an ontology’s structure. Involves domain analysis, defining concepts and relations, and iteratively refining the model.
- Ontology Learning: Automatic extraction of ontology elements from text or data . For example,text mining techniques can induce new classes or relations (e.g. “Type of X”, hyponymy patterns) to populate or expand an ontology.
- Ontology Matching/Alignment: Finding correspondences between concepts in different ontologies. This is crucial when integrating data: e.g., aligning Customer in one ontology to a client in another.
- Ontology Mapping: Ontology mapping but unidirectional.
- Ontologies are also pruned or modified time to time, in order to ensure veracity against time.

Commonly Used Ontologies

- WordNet – Lexical ontology grouping words by meaning (synsets).
- DBpedia / Wikidata – Structured data from Wikipedia.
- Schema.org – Ontology for describing web content. (Semantic Web)
- ConceptNet – Commonsense knowledge graph.

All of the ontologies mentioned above are middle ontologies or general-pupose ontologies. The ones below are a superset of some of these ontologies, along with other ontologies, linking these ontologies for interoperability and to address words that overlap these ontologies.

- SUMO
- WordNet Top Ontology
- Cyc Upper Ontology

Tools for Ontology Building

- Ontology Editors: Software for creating/editing ontologies.

  E.g. Protégé, NeOn Toolkit, OWLGrEd, etc.
  These provide GUI support for defining classes, properties, instances, and visualizing the ontology graph. Most linguists are experts from non-stem backgrounds usually use Protégé with Manchester OWL. Python users can also use libraries like : `owlready2`, `rdflib`, and `OntoSpy`

- RDF(S) and OWL: Standard languages/formats for ontologies.

  RDF Schema (Resource Description Framework Schema) defines basic class/property hierarchies; OWL (Web Ontology Language) (in DL, EL, QL profiles) provides
  rich expressivity.

Applications

- SEO and Semantic Search Engines (ranked results and responses).
- Chatbots and Virtual Assistants (context-aware responses).
- Machine Translation (meaning-preserving translation).
- Information Extraction (from unstructured text).
- Knowledge Graphs (Schema.org, DBpedia, Wikidata)
- Cross-Domain Reasoning: By linking ontologies (e.g. SUMO, DOLCE) with domain data, systems can perform cross-domain queries (e.g. finding environmental impacts of chemical processes by reasoning over a linked ontology.

Examples

- *https://schema.org/Person* → *Example of how people should be described in an ontology, aimed for web-devs.*
- *https://ontobee.org/* → *A searchable engine for biomedical ontologies.*
- *http://aber-owl.net/* →*Same as previous, but slow and not really that resourceful*